Prob. 3.

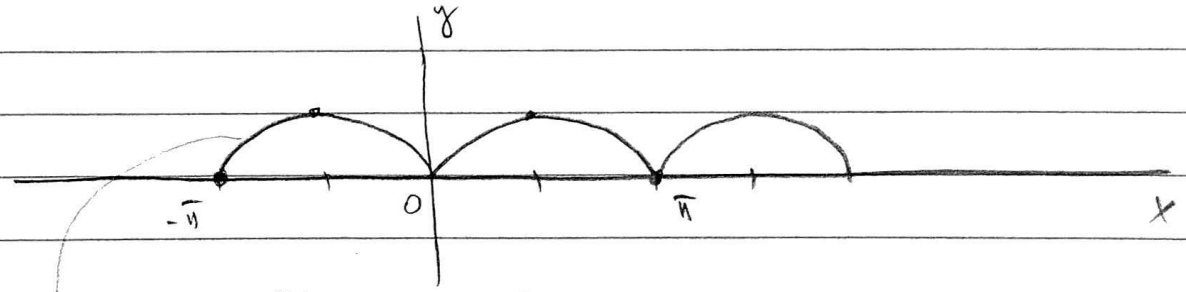
Derive the following F.S. repr-n, & sketch the f-n to which it converges:

(1) $$\underbrace{\sin(x)}_{f(x)} = \underbrace{\frac{2}{\pi} - \frac{4}{\pi} \cdot \sum_{k=1}^{\infty} \frac{1}{4k^2-1} \cos(2kx)}_{F.S.}, \quad 0 \le x \le \pi.$$

Sol-n.

Since $\sin(x)$ is an odd f-n, while the RHS of (1), F.S, is an even f-n, LHS & RHS can not equal each other on $[-\pi, \pi]$, but they can equal each other on $[0, \pi]$. Hence let us <u>extend evenly</u> our f-n $\sin(x)$ defined on $[0, \pi]$ to a f-n on $[-\pi, \pi]$:

$g(x)$:



$$g(x) = \begin{cases} \sin(x), & [0, \pi] \\ -\sin(x) & [-\pi, 0) \end{cases}$$

$g(x)$ = even extension of $f(x) = \sin(x)$ on $[-\pi, \pi]$.

By the previous HW, we know how to find a_0, a_n, b_n, when f-n is odd or even on $[-\pi, \pi]$. Since our extended f-n $g(x)$ is even on $[-\pi, \pi]$, the coef-ts are:

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} \overset{g \text{ is even}}{g(x)}\, dx = \frac{2}{\pi} \int_0^{\pi} \sin(x)\, dx = \frac{2}{\pi} \left[\cos(x) \right]_0^{\pi} =$$

(2) $$= \frac{2}{\pi} [-1-1] = \frac{-4}{\pi}. \quad (3) \quad \boxed{a_0 = \frac{-4}{\pi}}$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \overbrace{\overbrace{g(x)}^{\text{even}}\,\overbrace{\cos(nx)}^{\text{even}}}^{\text{even}}\, dx = \frac{2}{\pi}\int_{0}^{\pi} \sin(x)\cdot\cos(nx)\, dx$$

$$= \frac{2}{\pi}\int_{0}^{\pi} \frac{1}{2}\left\{\sin\left[(1-n)x\right] + \sin\left[(1+n)x\right]\right\} dx =$$

$$= \frac{1}{\pi}\int_{0}^{\pi} \sin\left[(1-n)x\right] + \sin\left[(1+n)x\right]\, dx =$$

$$= \frac{1}{\pi}\left\{-\frac{\cos\left[(1-n)x\right]}{(1-n)} + \frac{-\cos\left[(1+n)x\right]}{(1+n)}\right\}_0^{\pi} =$$

$$= \frac{1}{\pi}\cdot\left\{-\frac{\cos\left[(1-n)\pi\right]}{1-n} + \frac{\cos\left[(1-n)0\right]}{1-n} - \frac{\cos\left[(1+n)\cdot\pi\right]}{1+n} + \frac{\cos\left[(1+n)\cdot 0\right]}{1+n}\right\}$$

$$= \frac{1}{\pi}\left\{-\frac{1}{n-1} + \frac{1}{n+1} + \frac{\cos\left[(n-1)\pi\right]}{n-1} - \frac{\cos\left[(n+1)\pi\right]}{n+1}\right\}, \Rightarrow$$

(5) $$\boxed{a_n = \frac{1}{\pi}\left\{\frac{\cos\left[(n-1)\pi\right]}{n-1} - \frac{\cos\left[(n+1)\pi\right]}{n+1} + \frac{1}{n+1} - \frac{1}{n-1}\right\}} \quad n = 1, 2, \ldots$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \underbrace{\overbrace{g(x)}^{\text{even}}\cdot\overbrace{\sin(nx)}^{\text{odd}}}_{\text{odd}}\, dx = \frac{1}{\pi}\int_{-\pi}^{\pi} \text{odd}(x)\, dx = 0.$$

(6) $$\boxed{b_n = 0} \quad n = 1, 2, \ldots$$

So the F. S. for the extended f-n $g(x)$ is as follows:

(7) $$g(x) = \frac{1}{2}\cdot\left(\frac{-4}{\pi}\right) + \sum_{n=1}^{\infty} \frac{1}{\pi}\left\{\frac{\cos[(n-1)\pi]}{n-1} - \frac{\cos[(n+1)\pi]}{n+1} + \frac{1}{n+1} - \frac{1}{n-1}\right\}\cos(nx), \quad [-\pi,\pi]$$

Let $\boxed{n = 2k-1}$, $k = 1, 2, \ldots$

$$a_{2k-1} = \frac{1}{\pi}\left\{\frac{\overbrace{\cos[(2k-2)\pi]}^{1}}{(2k-2)} - \frac{\overbrace{\cos[(2k)\pi]}^{1}}{(2k)} + \frac{1}{(2k)} - \frac{1}{(2k-2)}\right\}\cos[(2k-1)x]$$

$$= \frac{1}{\pi}\left\{\frac{1}{(2k-2)} - \frac{1}{(2k)} + \frac{1}{(2k)} - \frac{1}{(2k-2)}\right\}\cos[(2k-1)x] = 0$$

So

(8) $$\boxed{a_{(2k-1)} = 0} \qquad \text{for } k = 1, 2, \ldots$$

Let

$\boxed{n = 2k}$

$$a_{2k} = \frac{1}{\pi}\left\{\frac{\overbrace{\cos[(2k-1)\pi]}^{=-1}}{(2k-1)} - \frac{\overbrace{\cos[(2k+1)\pi]}^{=-1}}{(2k+1)} + \frac{1}{(2k+1)} - \frac{1}{(2k-1)}\right\}\cdot\cos(2kx)$$

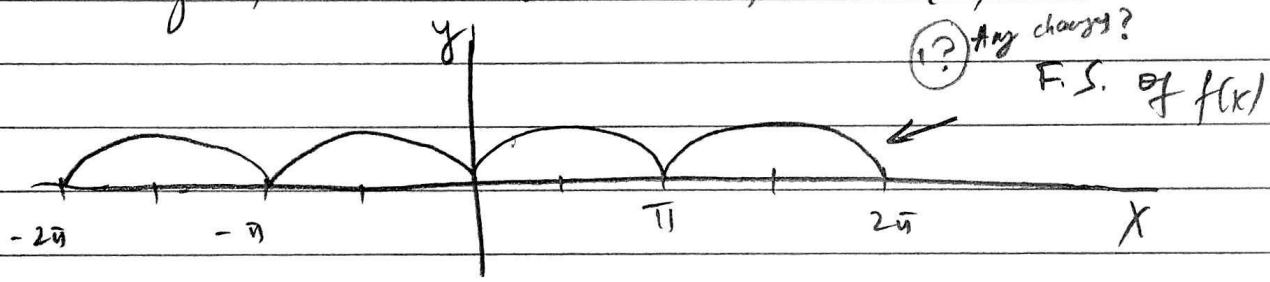
$$= \frac{1}{\pi}\left\{-\frac{1}{(2k-1)} + \frac{1}{(2k+1)} + \frac{1}{(2k+1)} - \frac{1}{(2k-1)}\right\}\cdot\cos(2kx) =$$

$$= \frac{1}{\pi}\left\{\frac{2^{\backslash 2k-1}}{2k+1} - \frac{2^{\backslash 2k+1}}{2k-1}\right\}\cos(2kx) = \frac{2}{\pi}\cdot\frac{(-2)}{4k^2-1}\cdot\cos(2kx), \Rightarrow$$

(9) $$\boxed{a_{2k} = \frac{-4}{\pi}\cdot\frac{1}{4k^2-1}\cdot\cos(2kx)} \qquad \text{for } k = 1, 2, \ldots$$

Putting (8) & (9) into seven by restricting $g(x) = \sin(x)$ on $[0, \pi]$, then we have

(10) $$\sin(x) = -\frac{2}{\pi} + \sum_{k=1}^{\infty} -\frac{4}{\pi} \cdot \frac{1}{4k^2-1} \cdot \cos(2k\,x) =$$

$$= -\frac{2}{\pi} - \frac{4}{\pi} \cdot \sum_{k=1}^{\infty} \frac{1}{4k^2-1} \cdot \cos(2k\,x)$$

establishing the desired result in (1).